Homework 7

August 15, 2014

Problem 2

(a) Let $g_n = f_n - f$ g_n continuous, decreasing, and converges to 0 pointwise.

See g_n converges to 0 uniformly on [a,b].

Given $\epsilon > 0$, let K_n be the set of all $x \in [a, b]$ with $g_n(x) \geq \epsilon$.

K_n is the preimage of a continuous function (g_n) on a closed set, so it is closed and compact (because it is a closed subset of $\mathbb{R}$).

g_n decreasing $\Rightarrow K_n \supset K_{n+1}$

Fix $x \in K$. Since, $g_n(x) \to 0$ which is less than ϵ, we see $x \notin K_n$ for sufficiently large n. Which means $x \notin \cap K_n$ and $\cap K_n = \emptyset$.

K_N is empty for some N (by compactness). $\Rightarrow 0 \leq g_n(x) < \epsilon \ \forall x \in \{K_n\}$ and $n \geq N$.

$\Rightarrow f_n$ converges uniformly on [a,b].

(b) Let $\epsilon > 0$ be given.

Let $g_n = f - f_n$ and K_n be the set of $x \in [a, b]$ such that $g_n(x) < \epsilon$. K_n is the preimage of a continuous function over an open set, so it is open.

f_n is increasing monotonically, so g_n is decreasing and K_n is increasing ($K_{n+1} \supset K_n$).

$f_n \to f$ pointwise, so $\{K_n\}$ is an open cover of [a,b] (by compactness). Since $K_{n+1} \supset K_n$ and $\{K_n\}$ is an open cover of [a,b], $\exists$ N s.t. $K_N = [a, b]$.

In other words, for $n > N$, $|f_n(x) - f(x)| < \epsilon \ \forall x \in [a, b]$. Or $f_n(x)$ converges uniformly to f on [a,b].

(c) See $\sum \frac{x^n}{1+x^n}$ converges for $x \in [0,1)$ by the root test.

$\lim_{n\to\infty} \sqrt[n]{\frac{x^n}{1+x^n}} = \lim_{n\to\infty} \frac{x}{\sqrt[n]{1+x^n}} < 1$ for $x \in [0,1)$

$\sum \frac{x^n}{1+x^n}$ converges on [0, 1).

(d) The sequence $\frac{x^n}{1+x^n}$ is decreasing monotonically on [0, a] for any $a \in [0,1)$.

$\lim_{n\to\infty} \frac{x^n}{1+x^n} = 0$ for $x \in [0,1)$, so $\frac{x^n}{1+x^n} \to 0$ pointwise.

By (a) $\frac{x^n}{1+x^n}$ converges uniformly on [0, a].

(e) $|f - f_n| = \sum \frac{x^n}{1+x^n}|$. See for any n, there exists x such that $|f - f_n| \geq \frac{3}{4}$. $x^n \geq 3 \Rightarrow \sum \frac{x^n}{1+x^n} \geq \frac{3}{4} \Rightarrow \sum \frac{x^n}{1+x^n} \geq \frac{3}{4}$ for $x \geq \sqrt[n]{3}$. Contradiction to the definition of uniform continuity, so $\sum \frac{x^n}{1+x^n}$ is not uniformly convergent.